

Arghamitra Talukder

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EDUCATION

2023 - 2028	PhD in Computer Science, Columbia University <i>NSF CSGrad4US Fellow</i>	New York, NY GPA: 3.98/4.0
2023 - 2025	MS in Computer Science (en route)	
2018 - 2021	BS in Electrical Engineering at Texas A&M University <i>Engineering Honors, Magna Cum Laude</i>	College Station, TX GPA: 3.78/4.0

PROJECTS

Predicting Exon Inclusion Levels via Contrastive Sequence Models

Developed a deep learning model to analyze gene regulation with a contrastive learning framework. This model identifies conserved functional elements by mapping evolutionarily related DNA sequences into a shared latent space, providing novel insights into gene expression.

Bayesian hierarchical isoform quantification with hybrid sequencing

Developed JOLI, a novel hierarchical model that enhances RNA transcript quantification by jointly integrating short-read (SR) and long-read (LR) sequencing data. The model utilizes a multi-sample learning framework to improve estimation accuracy and reproducibility, particularly for low-abundance isoforms.

Modeling cell plasticity with integrated single-cell data

Designed a statistical model to quantify cellular plasticity by integrating single-cell lineage and gene expression data. This approach utilizes a continuous annotation distribution methodology to statistically detect and score plastic cell states.

PUBLICATIONS

Talukder, Arghamitra, Nicholas Keung, et al. (2026). “CLADES — Contrastive Learning Augmented Differential Splicing with Orthologous Positive Pairs”. In: *RECOMB (Accepted)*. [Online version](#).

Talukder, Arghamitra, Shree Thavarekere, et al. (2025). “Multi-sample, multi-platform isoform quantification using empirical Bayes”. In: *RECOMB-Seq (Short-talk)*. [Online version](#).

Talukder, Arghamitra and Itsik Pe’er (2024). “Computational Method to Detect Cancerous Plasticity”. In: *IICD Workshop on Single-Cell Data Integration & OT, CRA-WP Grad Cohort for IDEALS (Short-talk)*.

Talukder, Arghamitra, Rujie Yin, et al. (2022). “Does inter-protein contact prediction benefit from multi-modal data and auxiliary tasks?” In: *MLSB, NeurIPS*. [Online version](#).

Ibrahim, Bassem, **Talukder, Arghamitra**, and Roozbeh Jafari (2020). “Multi-source multi-frequency bio-impedance measurement method for localized pulse wave monitoring”. In: *EMBC, IEEE*. [Online version](#).

WORK EXPERIENCE

Product & Test Engineer at Texas Instruments, Dallas June 2021 - Aug 2023

- Responsible for TI customized Ultra-High Voltage test technology with 70% cost efficiency.
- Created a product specific test program converting from legacy tester to the latest technology reducing 40% of the test time.

Test & Validation Intern at Texas Instruments, Dallas June 2020 - Aug 2020

- Enabled traceability with the accuracy of 99.9% by device ID recovery obtained using OCR technology.
- Automation of data acquisition and validation by database access and scripting using SQL and PERL.

FELLOWSHIPS

NIH T15 Training Grant	Sep 2025 - Aug 2026
CISE CSGrad4US Fellowship, CRA, NSF	Sep 2023 - Aug 2027
NeurIPS Travel Award	Dec 2022

TEACHING ASSISTANT

Machine Learning for Functional Genomics	Sep 2025 - Dec 2025
Computational Genomics	Jan 2025 - May 2025
Experimental Physics Lab II-Mechanics	Jan 2019 - Dec 2019
Foundation of Engineering in Python and LabVIEW	Sep 2018 - Dec 2018

SKILLS

Programming & ML	Python, PyTorch, PyTorch Lightning, MATLAB, Perl, C, Bash
Genomics & Data	Genomic sequence processing, RNA-seq (short- and long-read)
Systems & Cloud	High-performance computing (SLURM, multi-GPU clusters), distributed training, Linux environments
Engineering & Testing	Python/Perl/C-based semiconductor test program development, legacy tester migration and test time reduction
Tools & DevOps	Git, Hydra/OmegaConf, WandB, Docker (basic)